COMPUTER COMMUNICATIONS & NETWORKS — QUIZ 2

Module: Lecture 2 Part 2 (Network Topologies, Physical Layer & Sine Wave Signal Analysis)

Name: _____ Student ID: _____ Score: _____ / 20

1. Which of the following is a major operational DISADVANTAGE of a full Mesh network topology?

- ◯ a. A central switch failure will cause the entire network to immediately stop working.
- ◯ b. An extraordinarily large amount of cabling and numerous I/O ports are required per device, making it expensive, physically bulky, and difficult to scale.
- ◯ c. If any single cable gets severed, the entire network shuts down.
- ◯ d. All communicating devices are forced to share a single collision domain and frequency channel.
- ◯ e. It cannot transmit digital signals and works only for analog audio lines.

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2. What is the critical vulnerability inherent to a standard, unidirectional Ring topology that can disrupt communications across the entire network?

- ◯ a. The central hub requires periodic replacement due to overheating.
- ◯ b. A failure or physical disconnection of any single node or cable segment breaks the token loop and collapses the entire network.
- ◯ c. It can strictly accommodate a maximum of three connected nodes.
- ◯ d. Data packets travel without parity checks or checksum verification.
- ◯ e. It causes continuous data packet collisions because nodes lack media access control.

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3. In a Star network topology, what is the primary role of the central connecting device (such as a switch or active hub)?

- ◯ a. It dynamically assigns public IP addresses to every connected host interface.
- ◯ b. It permanently stores all transmitted application documents on a central hard disk.
- ◯ c. It acts as a central junction and forwarding hub that regulates traffic flow between devices and can regenerate/repeat degraded signals.
- ◯ d. It encrypts all traffic using public key infrastructure before forwarding frames.
- ◯ e. It converts all incoming digital frames into modulated analog microwave signals.

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4. A startup office requires a cost-effective local network where a single device or cable disconnection does NOT take down other users, and new workstations can be added or removed easily without interrupting ongoing work. Which topology BEST meets these operational criteria?

- ◯ a. Ring topology
- ◯ b. Bus topology
- ◯ c. Mesh topology
- ◯ d. Star topology
- ◯ e. Point-to-point simplex line

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5. In a legacy Bus topology, all workstations connect directly to a single shared backbone cable. What happens if this central backbone trunk cable encounters a physical break or cut?

- ◯ a. Only the specific workstation closest to the cut is disconnected.
- ◯ b. The workstations automatically activate built-in mesh failovers.
- ◯ c. The entire network collapses because signal reflection and loop termination failure halt all communication across the bus.
- ◯ d. Data packets are automatically rerouted through the central switch.
- ◯ e. The network divides into two independent wireless subnets without packet loss.

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6. Which statement accurately captures the distinction between *data* and a *signal* in communication systems?

- ◯ a. Data is the abstract representation of human/computational information; a signal is the measurable physical electromagnetic wave or energy variation that travels over a medium.
- ◯ b. Data travels through air, whereas signals travel only through copper cables.
- ◯ c. A signal is stored in persistent databases; data is the voltage that dissipates immediately.
- ◯ d. Data and signal are completely synonymous and interchangeable terms for binary bits.
- ◯ e. Data is always analog; signals are always digital.

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7. What is the fundamental responsibility of the Physical Layer in the OSI reference model?

- ◯ a. Deliver packets across multiple interconnected routing networks.
- ◯ b. Move raw bits across a physical communication link by transforming them into appropriate physical signals.
- ◯ c. Provide end-to-end flow control and ensure retransmission of dropped segments.
- ◯ d. Parse syntax, character formatting, and encryption for application protocols.
- ◯ e. Establish and terminate software dialogues between operating systems.

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8. The Physical Layer represents digital bits using physical signals propagated across transmission media. Which set accurately lists the primary physical signal types utilized?

- ◯ a. IP packets, MAC frames, and TCP segments
- ◯ b. Electrical voltage/current signals (copper), light pulses (optical fiber), and electromagnetic radio/microwave waves (wireless)
- ◯ c. ASCII characters, Unicode values, and binary integer codes
- ◯ d. HTML markup tags, XML trees, and JSON objects
- ◯ e. Hexadecimal addresses, decimal port identifiers, and socket numbers

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9. An analog sine wave signal has a peak positive voltage reaching $+5\ \text{V}$ and a lowest negative voltage reaching $-5\ \text{V}$. It takes exactly $2.0\ \text{seconds}$ to complete one full cycle. At time $t = 0\ \text{s}$, the signal starts at its peak value ($y(0) = +5\ \text{V}$). Determine the amplitude (A), frequency (f), and phase angle (ϕ) in degrees ($0^\circ - 360^\circ$).

Formulas: $f = \frac{1}{T}$; General Form: $y(t) = A\sin(2\pi ft + \phi)$

Write your answers:

$A =$ _____ V, $f =$ _____ Hz, $\phi =$ _____ degrees

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10. A periodic sine wave completes exactly 3 complete cycles in $1.5\ \text{seconds}$. Its peak value is $4\ \text{V}$. At time $t = 0\ \text{s}$, its instantaneous value is at the lowest negative peak ($y(0) = -4\ \text{V}$). Which mathematical equation correctly models this signal over time?

- ○ a. $y(t) = 4\sin(2\pi t + \frac{\pi}{2})$
- ○ b. $y(t) = 4\sin(4\pi t - \frac{\pi}{2})$
- ○ c. $y(t) = 4\sin(4\pi t + \frac{\pi}{2})$
- ○ d. $y(t) = 4\sin(2\pi t - \frac{\pi}{2})$
- ○ e. $y(t) = 4\sin(6\pi t - \frac{\pi}{2})$

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11. A periodic carrier signal operates at a frequency of $f = 250\ \text{Hz}$. What is the exact period (T) of this signal expressed in milliseconds (ms)?

Formula: $T = \frac{1}{f}$

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12. A transmitted signal is governed by the equation:

$$s(t) = 12\sin\left(100\pi t + \frac{\pi}{4}\right)$$

Determine the signal's peak amplitude (A), frequency (f in Hz), and phase shift (ϕ in degrees).

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13. An organization connects 8 core datacenter routers together in a complete full-mesh topology. Calculate the total number of physical duplex cables required to interconnect every router directly to every other router.

$$\text{Links} = \frac{n(n-1)}{2}$$

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14. How does a Dual-Ring topology (such as Fiber Distributed Data Interface - FDDI) overcome the single-point-of-failure disadvantage of a standard simple ring?

- ◯ a. It replaces all fiber cables with wireless links.
- ◯ b. It uses a secondary, counter-rotating ring that automatically wraps around a severed cable or broken node to restore an intact loop.
- ◯ c. It introduces a central switch that converts the entire network into a bus.
- ◯ d. It doubles the speed of light in the optical glass.
- ◯ e. It discards token passing in favor of random broadcasting.

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15. If a sine wave with period T is shifted forward in time by exactly one-quarter of its cycle ($\Delta t = \frac{T}{4}$), what is the corresponding phase shift in radians and degrees?

- ◯ a. $\frac{\pi}{4}$ rad (45°)
- ◯ b. $\frac{\pi}{2}$ rad (90°)
- ◯ c. π rad (180°)
- ◯ d. $\frac{3\pi}{2}$ rad (270°)
- ◯ e. 2π rad (360°)

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16. Which of the following fundamentally differentiates an *analog* signal from a *digital* signal?

- ◯ a. An analog signal can only be transmitted over copper, whereas digital travels only via lasers.
- ◯ b. An analog signal varies smoothly and continuously over an infinite continuum of values, whereas a digital signal consists of discrete, finite levels (e.g., 0 and 1).
- ◯ c. Digital signals do not experience attenuation or noise.
- ◯ d. Analog signals have no frequency or wavelength.
- ◯ e. Digital signals are immune to bandwidth constraints.

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17. An electrical alternating current signal has a standard mains frequency of $f = 50$ Hz. What is the angular frequency (ω) of this signal in radians per second?

Formula: $\omega = 2\pi f$

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18. A wireless access point transmits electromagnetic signals at a carrier frequency of $f = 2.4$ GHz (2.4×10^9 Hz). Assuming the propagation speed of light in air is $c = 3.0 \times 10^8$ m/s, calculate the wavelength (λ) of the signal.

$$\lambda = \frac{c}{f}$$

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19. In a Star network topology comprising $n = 20$ workstations connected to an enterprise Ethernet switch, how many dedicated physical communication links and device network interface ports are required in total?

- ◯ a. 190 links, 380 total ports
- ◯ b. 20 links, 20 workstation ports + 20 switch ports (40 ports total)
- ◯ c. 1 single shared link, 20 ports
- ◯ d. 10 links, 20 ports total
- ◯ e. 40 links, 20 ports total

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20. What is a *Hybrid Topology*, and why is it commonly deployed in enterprise campus networks?

- ◯ a. A network that mixes optical fiber with wireless links while using no routers.
- ◯ b. A network topology combining two or more distinct basic topologies (such as Star-Bus or Tree) to leverage individual advantages and optimize scalability.
- ◯ c. A network operating without standard protocols or addressing.
- ◯ d. A network that requires all computers to be identical in brand and operating system.
- ◯ e. A network where all packets travel in reverse order.

ANSWER KEY & DETAILED EXPLANATIONS

Lecture 2 Part 2 — Solutions

Question 1: b Mesh Topology Drawbacks

Explanation: In a full mesh network, connecting n devices requires $\frac{n(n-1)}{2}$ links, and each individual device must be equipped with $n-1$ dedicated I/O ports. This creates severe installation complexity, extreme cabling volume, high hardware costs, and very poor physical scalability.

Question 2: b Ring Vulnerability

Explanation: In a standard ring topology, data circulates sequentially through each intermediate station that functions as a repeater. If any single workstation is turned off, disconnected, or if a single cable breaks, the closed loop is fractured and all communication halts.

Question 3: c Star Central Junction

Explanation: In a star topology, end nodes do not connect directly to one another; each node has a dedicated point-to-point link to the central hub or switch. The central node acts as the traffic junction that manages, forwards, and repeats data streams.

Question 4: d Topology Selection

Explanation: Star topology is the industry standard for modern enterprise office LANs because it isolates node failures: if a single workstation or cable fails, only that device is affected, leaving the rest of the network operational. Adding or relocating nodes is trivially managed at the central patch panel/switch.

Question 5: c Bus Cable Cut

Explanation: In a bus topology, signals propagate across the shared trunk wire and must be absorbed by terminators at both physical ends. A break in the trunk splits the cable into unterminated stubs, causing severe electrical signal reflection that corrupts all transmissions and halts the entire network.

Question 6: a Data vs Signal

Explanation: *Data* refers to the conceptual information entities (text, numbers, audio, pixels) that humans or computers understand. A *signal* is the physical, measurable transmission of energy (electrical voltage, optical pulses, radio electromagnetic waves) propagated over a medium to represent that data.

Question 7: b Physical Layer Role

Explanation: The primary duty of Layer 1 (Physical Layer) is to receive a stream of bits from Layer 2 (Data Link) and encode/modulate those bits into physical signals suitable for transmission across the physical transmission medium.

Question 8: b Signal Types

Explanation: Physical Layer transmission relies on three primary signal modalities: electrical currents/voltages along copper cables (e.g., UTP/coaxial), pulses of light down glass/plastic optical fiber, and radiated electromagnetic radio/microwave waves propagated through the air.

Question 9: A = 5 V, f = 0.5 Hz, phase = 90 degrees Sine Wave Parameters

Explanation:

- **Amplitude (A):** Peak deviation from zero baseline is $+5\ \text{V}$.
- **Period (T):** The signal completes one full cycle in $2.0\ \text{s}$. Therefore:

$$f = \frac{1}{T} = \frac{1}{2.0\ \text{s}} = 0.5\ \text{Hz}$$

- **Phase (ϕ):** At $t = 0$, $y(0) = 5\ \text{V}$. Since $y(t) = A\sin(2\pi ft + \phi)$, $5 = 5\sin(\phi) \implies \sin(\phi) = 1 \implies \phi = 90^\circ$ (or $\frac{\pi}{2}$ rad).

Question 10: b Sine Wave Equation

Explanation: The wave completes 3 cycles in $1.5\ \text{s}$, so:

$$f = \frac{3}{1.5\ \text{s}} = 2\ \text{Hz}$$

Angular frequency:

$$\omega = 2\pi f = 2\pi(2) = 4\pi\ \text{rad/s}$$

Peak amplitude $A = 4\ \text{V}$. At $t = 0$, the wave is at $-4\ \text{V}$:

$$y(0) = 4\sin(\phi) = -4 \implies \sin(\phi) = -1 \implies \phi = -\frac{\pi}{2}\ \text{rad}\ (\text{or } 270^\circ)$$

Substituting into $y(t) = A\sin(\omega t + \phi)$ gives:

$$y(t) = 4\sin\left(4\pi t - \frac{\pi}{2}\right)$$

Question 11: 4 ms Period Calculation

Explanation:

$$T = \frac{1}{f} = \frac{1}{250\ \text{Hz}} = 0.004\ \text{s}$$

Converting to milliseconds ($1\ \text{s} = 1,000\ \text{ms}$):

$$T = 0.004 \times 1,000 = 4\ \text{ms}$$

Question 12: A = 12 V, f = 50 Hz, phase = 45 degrees Signal Parameter Extraction

Explanation: From standard form $s(t) = A\sin(2\pi ft + \phi)$:

- **Amplitude (A):** $12\ \text{V}$
- **Frequency (f):** $2\pi f = 100\pi \implies f = \frac{100\pi}{2\pi} = 50\ \text{Hz}$
- **Phase (ϕ):** $\phi = \frac{\pi}{4}\ \text{rad} = \left(\frac{180^\circ}{\pi}\right) \times \frac{\pi}{4} = 45^\circ$

Question 13: 28 links Mesh Formula

Explanation:

$$\text{Links} = \frac{n(n-1)}{2} = \frac{8(8-1)}{2} = \frac{8 \times 7}{2} = \frac{56}{2} = 28\ \text{links}$$

Question 14: b Dual Ring Redundancy

Explanation: In a dual-ring architecture (e.g., FDDI or Sonet/SDH rings), two concentric loops carry data in opposing directions. If a cable is severed or a node fails, the stations adjacent to the break isolate the faulted segment and bridge primary traffic onto the secondary ring, re-establishing a closed operational circuit.

Question 15: b Phase Angle

Explanation: One full cycle corresponds to 360° or 2π radians of phase. Shifting by one-quarter cycle corresponds to:

$$\phi = \frac{1}{4} \times 360^\circ = 90^\circ \quad \left(\text{or } \frac{1}{4} \times 2\pi = \frac{\pi}{2} \text{ rad} \right)$$

Question 16: b Analog vs Digital

Explanation: Analog signals vary continuously with time, having infinitely many possible amplitude levels within a specified range. Digital signals vary in discrete, quantified steps, representing information with a finite set of discrete states (most commonly two binary voltage levels).

Question 17: 314.16 rad/s (or 100pi rad/s) Angular Frequency

Explanation:

$$\omega = 2\pi f = 2 \times \pi \times 50 \text{ Hz} = 100\pi \text{ rad/s} \approx 314.16 \text{ rad/s}$$

Question 18: 0.125 m (or 12.5 cm) Wavelength Calculation

Explanation:

$$\lambda = \frac{c}{f} = \frac{3.0 \times 10^8 \text{ m/s}}{2.4 \times 10^9 \text{ Hz}} = \frac{3.0}{24} = 0.125 \text{ m} = 12.5 \text{ cm}$$

Question 19: b Star Topology Port Count

Explanation: In a star topology with n nodes, each node requires exactly one dedicated link to the central switch. Thus, there are $n = 20$ physical links. This requires 20 host network ports (1 per workstation) and 20 switch ports (20 switch interface ports), totaling 40 active I/O ports.

Question 20: b Hybrid Networks

Explanation: A hybrid topology merges two or more classic topologies (for example, connecting multiple Star networks together along a high-speed backbone, known as a Star-Bus or Tree topology). This combines the fault isolation of Star networks with the extended geographic reach and cost efficiencies of backbone networks.